

Inventory-Deus — Entity Reference

All 16 services, every entity, and what each one functions as.

1. Product / Catalog Service

Entity	Functions as
Product	The conceptual item ("Brand-X widget"). Holds name, description, brand, category. One product can have many SKUs.
SKU	The actual sellable unit ("Brand-X widget, red, 500ml"). The atomic identifier the rest of the system tracks quantities against.
Variant	An axis of variation (color, size, flavor). Defines how one Product splits into multiple SKUs.
Category	A bucket for grouping products ("Beverages > Soft Drinks"). Drives reporting and rules.
Attribute	Free-form key/value pairs (material, voltage, weight). Lets you search/filter without changing the schema.

2. Inventory Service

Entity	Functions as
StockLevel	"How many of SKU X are at Warehouse Y in Bin Z right now." The number people actually look at.
Location	A logical position — usually a (warehouse, zone, bin) tuple.
Bin	The smallest physical spot stock can sit.
Lot	A batch of identical units sharing an expiry/manufacture date (medicines, food).
Serial	A single trackable unit (electronics, firearms — every piece has its own ID).
ReservedQty	Stock that is physically present but already promised to a sales order or transfer — invisible to new orders.

3. Warehouse Service

Entity	Functions as
Warehouse	A physical building/branch. The top of the location hierarchy.
Zone	A subdivision inside a warehouse (Receiving, Bulk, Pick Face, Cold Storage).
Bin	An addressable storage slot (Aisle-Row-Shelf).
Capacity	How much a zone/bin can hold (cubic volume, weight, pallet count).
PickingStrategy	Rule for which stock to pick first — FIFO, LIFO, FEFO (expiry), nearest-bin.

4. Supplier Service

Entity	Functions as
Supplier	A vendor you buy from — legal name, tax info, status.
Contact	A person at the supplier (sales rep, AP clerk) with phone/email.
LeadTime	Days from PO to delivery for a given supplier+SKU. Drives reorder timing.
PriceList	What this supplier charges for which SKUs, valid from/to.
Terms	Payment terms (Net-30), incoterms (FOB, CIF), return policy.

5. Procurement / Purchase Service

Entity	Functions as
PurchaseOrder	The header document — supplier, dates, status (DRAFT/APPROVED/RECEIVED).
PO_Line	One row per SKU on the PO — qty, unit price, due date.
GRN	Goods Receipt Note — confirmation that goods physically arrived against a PO.
Invoice	The supplier's bill — matched against the PO and GRN ("3-way match").
Approval	The signoff trail — who approved what at which threshold.

6. Sales Order Service

Entity	Functions as
SalesOrder	Customer demand — customer, ship-to, status.
SO_Line	One row per SKU ordered — qty, price, requested date.
Allocation	"These specific units in this specific bin are earmarked for this order line."
Backorder	Demand that could not be allocated because stock ran out — queues for future supply.

7. Stock Movement Service

Entity	Functions as
Movement	The generic record of stock changing place/state. Every quantity change is a movement.
Transfer	Movement between warehouses (or between zones).
Adjustment	Quantity correction with no physical movement (damage, theft, recount).
ReasonCode	Why the movement happened — drives reporting and audit ("RECEIPT", "DAMAGE", "EXPIRY").

8. Receiving Service

Entity	Functions as
ASN	Advance Ship Notice — supplier's heads-up that goods are coming, lets you prep.
Receipt	The actual record of what arrived (vs what was promised).
Putaway	Decision/record of which bin received goods go into.
Discrepancy	The gap between ordered, shipped, and received — drives supplier disputes.
QA_Check	Inspection result before goods are accepted into available stock.

9. Fulfillment / Shipping Service

Entity	Functions as
PickList	The walk-route a picker follows to collect items for one or many orders.
PackSlip	What is in the box — included with the shipment.
Shipment	The physical parcel/truck leaving the warehouse.
Carrier	Who is moving it (DHL, FedEx, internal fleet).
Tracking	The waybill/AWB number + status pings from the carrier.

10. Replenishment Service

Entity	Functions as
ReorderPolicy	The rule per SKU+warehouse — when to reorder, how much.
Forecast	Predicted demand for the coming period — feeds the policy.
Suggestion	A draft purchase request the system generates for a buyer to approve.
Min/Max	The two thresholds: drop below Min → reorder up to Max.

11. Audit / Counting Service

Entity	Functions as
CountSession	A scheduled cycle count — which warehouse/zone, when, who counts.
Variance	The difference between counted qty and system qty.
Approval	Manager signoff for variances above tolerance.
Adjustment	The corrective movement that closes the variance.

12. UoM / Conversion Service

Entity	Functions as
Unit	A measurement — each, case, pallet, kg, liter.
Conversion	The factor between units (1 case = 24 eaches; 1 pallet = 40 cases).
Barcode	The scannable code that resolves to a SKU + unit (a case barcode vs an each barcode).

13. Notification Service

Entity	Functions as
Alert	A specific message instance ("Low stock on WGT-001").
Template	The reusable body with placeholders.
Channel	How it is delivered (email, SMS, in-app, webhook).
Subscriber	Who receives which alert types.

14. Auth / User / Role Service

Entity	Functions as
User	A login — email, password hash, status.
Role	A named bundle of permissions (Buyer, Picker, Manager).
Permission	A single allowed action ("po.approve", "stock.adjust").
WarehouseAccess	Which physical warehouses a user is allowed to operate in.

15. Reporting / Analytics Service

Entity	Functions as
Report	A defined query/output (Stock Aging, Turn Rate, Shrinkage).
Metric	A single computed KPI value ("Inventory Turns Q1 = 4.2").
Snapshot	A frozen point-in-time copy of stock for historical comparison.
Export	A generated file (Excel/PDF/CSV) ready to download.

16. Pricing Service

Entity	Functions as
PriceBook	A catalog of selling prices, scoped by customer tier / region / time.
Cost	Landed cost per SKU (purchase price + freight + duty).
Margin	Selling price minus cost — the profit-per-unit number.
DiscountRule	Conditional markdown ("10% off if qty > 100").

Headline summary

16 services, roughly 75 entities. Five are load-bearing — if these are airtight, everything else is decoration on top:

SKU	what the thing is
Warehouse / Bin	where it sits
StockLevel	how much is there
Movement	why the quantity changed
PurchaseOrder / SalesOrder	the supply and demand triggers